

Graph-Augmented Fine-Tuning of Open-Source Large Language Models for Legal Document Understanding

Encoding citation and cross-reference structure as a heterogeneous knowledge graph, and injecting it into an open-source language model as continuous soft tokens under a parameter-efficient fine-tuning budget.

Document scope

This document sets out the objective, the state of the art, the dataset plan, the system architecture, the experimental design and the evaluation protocol for the project. It is intended to be complete enough to serve as the basis for implementation and for a paper submission, and specific enough that every claim it makes is testable.

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Item	Value
Base models	Llama 3.1 8B Instruct, Qwen 2.5 7B, Mistral 7B v0.3, SaulLM-7B
Adaptation method	LoRA and QLoRA, rank 16 to 64, under 2 percent trainable parameters
Structural module	R-GCN or Heterogeneous Graph Transformer, 2 to 3 layers, plus MLP projector
Injection point	K continuous soft tokens prepended to the input embedding sequence
Task families	Judgment prediction, statute and precedent retrieval, clause extraction, grounded summarisation
Jurisdictions	United States, European Union and ECtHR, India
Compute envelope	One A100 80GB, or two RTX 4090 with 4-bit quantisation
Indicative duration	26 weeks to paper draft and artifact release

Bibliographic details in Section 13 should be verified against the publisher of record before external circulation.

1. Executive summary

Legal documents carry their meaning in a network of references. A judgment depends on the precedents it cites, the statutory provisions it applies, the court hierarchy that binds it and the amendment history that determines which version of a provision was in force on the relevant date. A contract clause depends on the definitions section it draws from, the cross-referenced clauses that qualify it and the governing law that constrains it. This structure is written down explicitly, in the form of citations and cross-references, and it is precisely what a standard autoregressive language model discards when it flattens a document into a token sequence.

This project proposes to keep that structure. Legal corpora are parsed into a heterogeneous knowledge graph with typed nodes and typed relations. For any input, a bounded and connected subgraph is retrieved, encoded with a relation-aware graph neural network, and projected into the token embedding space of an open-source language model as a small number of continuous soft tokens. The base model stays frozen and quantised; only the graph encoder, the projector and low-rank adapters are trained. The comparison that matters is against text-only parameter-efficient fine-tuning under an identical trainable-parameter budget.

The contribution is not the graph and not the fusion technique taken separately, since both exist. It is the intersection, which is currently untested. Legal NLP uses citation graphs with encoder-only classifiers that cannot generate or explain. The graph-plus-language-model literature has the fusion machinery but evaluates it on generic benchmarks whose graphs lack the properties that make legal structure interesting, namely an authority hierarchy, temporal validity windows on provisions, and asymmetric relations such as overruling and distinguishing.

Three outcomes are targeted and each is measurable. First, accuracy gains over text-only adaptation across four task families and three jurisdictions. Second, a reduction in citation hallucination, measured separately as validity, meaning the citation resolves to a real node, and as relevance, meaning it is actually pertinent, since reporting only the first is easy and misleading. Third, improved cross-jurisdiction transfer, tested by training on United States or European material and adapting to Indian law with limited target data.

2. Problem statement

2.1 Legal text is relational, and tokenisation destroys the relations

A citation such as a reference to section 302 of the Indian Penal Code, or to a reported case in a particular volume, is a pointer to a specific object in a specific corpus. To a tokeniser it is a short string of subword units that co-occurs with certain other strings. The model can learn the statistical shape of the pointer, which is why generated citations look plausible, but nothing in the objective requires the pointer to resolve to anything real. The structure is present in the data and absent from the representation.

2.2 Three failure modes follow

- **Citation hallucination.** Instruction-tuned models emit statutory references and case names that are syntactically well formed and factually nonexistent. Documented sanctions against practitioners who filed such material make this the most consequential failure mode in the domain, and it follows directly from the absence of any grounding between a generated citation and a verifiable corpus entry.
- **Context truncation and dilution.** Judgments routinely exceed 30,000 tokens and merger agreements exceed 100,000. Long-context models make this technically possible but not cheap or reliable, and attention quality degrades over the middle of very long inputs. Meanwhile the information that determines the answer is often a handful of paragraphs plus two or three cited authorities that live in entirely different documents, which no amount of context length on the current document will supply.
- **Weak jurisdictional transfer.** A model fine-tuned on United States case law does not carry over cleanly to Indian or European reasoning, because drafting convention, citation format, procedural structure and

doctrinal vocabulary all differ. Retraining from scratch per jurisdiction is not affordable for most research groups, so a representation that transfers is worth more than a model that is excellent in one system.

2.3 Why a graph, and why soft tokens

A graph is the natural representation because the relations are typed, directed and already written down. It is not an inferred latent structure that has to be discovered; it is an extraction problem with measurable precision. Soft tokens are the natural injection point because they cost a fixed and tiny number of positions regardless of subgraph size, whereas serialising a subgraph into the prompt costs tokens linear in the number of edges and forces the model to reconstruct from prose what it was handed as structure. Serialisation also degrades sharply as graphs grow, which is the empirical finding that motivates the whole soft token line of work.

3. Objectives, research questions and hypotheses

3.1 Primary objective

To design, implement and empirically validate a graph-augmented parameter-efficient fine-tuning framework that injects structural information from a legal knowledge graph into an open-source large language model, and to demonstrate that this yields measurable improvements over text-only fine-tuning on legal document understanding tasks under an identical trainable-parameter budget.

3.2 Specific objectives

#	Objective	Verifiable outcome
O1	Construct a heterogeneous legal knowledge graph from statutory corpora, case law and contracts, with typed nodes and typed relations	Released graph with node and edge counts reported; extraction precision and recall measured against a manually annotated sample of at least 500 references
O2	Design a query-conditioned subgraph retrieval mechanism that selects a bounded, connected, relevant subgraph for any input	Retrieval recall at a fixed node budget, compared against dense passage retrieval and BM25 baselines
O3	Train a heterogeneous graph encoder and cross-modal projector mapping subgraph structure into the token embedding space	Convergence of the alignment objective; ablation demonstrating that graph tokens carry non-trivial signal
O4	Fine-tune open-source LLMs in the 7B to 8B range using LoRA or QLoRA with graph soft tokens on a multi-task legal instruction mixture	Trained checkpoints and adapter weights, with trainable parameters held under 2 percent of total
O5	Benchmark against text-only PEFT, linearisation and retrieval-augmented baselines across four task families	Macro-F1, nDCG, ROUGE-L and citation faithfulness reported on held-out splits with significance testing
O6	Quantify the reduction in citation hallucination attributable to graph grounding	Hallucinated citation rate by automatic corpus verification, plus expert adjudication on a stratified sample
O7	Evaluate cross-jurisdiction transfer from a source legal system to a target with limited data	Few-shot transfer curves from a US or EU source to an Indian target at 0, 100, 1000 and 10000 target examples

3.3 Research questions

- **RQ1.** Does explicit structural encoding improve legal document understanding over text-only adaptation when the trainable-parameter budget is held constant?
- **RQ2.** Does learned graph encoding outperform serialising the same subgraph into the prompt as text, and how does the margin behave as subgraph size grows?
- **RQ3.** Does graph grounding reduce citation hallucination, and does the reduction hold for relevance as well as for bare validity?
- **RQ4.** Does structural knowledge transfer across jurisdictions more readily than lexical and doctrinal knowledge, given that citation topology is broadly similar across common law systems while vocabulary is not?

3.4 Hypotheses

ID	Hypothesis	Falsified if
H1	Graph soft tokens improve macro-F1 over text-only LoRA on citation-dependent tasks, with the largest margin on statute identification and precedent reasoning	No statistically significant gain over baseline B2 at matched budget across three seeds
H2	Learned graph encoding beats prompt linearisation, and the gap widens with subgraph size	Linearisation matches or exceeds the learned encoder at N greater than 50 nodes
H3	Structural grounding reduces citation relevance error, not merely validity error	Validity improves while expert-adjudicated relevance is unchanged, which would indicate the model has learned to emit common real citations rather than correct ones

3.5 Explicit non-objectives

The system is not intended to provide legal advice, to substitute for professional judgment, or to be deployed in any capacity affecting a real party. All evaluation is retrospective and benchmark-based. Outputs are framed as document-understanding predictions rather than legal conclusions. No claim is made about calibration sufficient for practice, and no deployment pathway is proposed within the scope of this document.

4. State of the art

4.1 Domain-adapted legal language models

The encoder generation established the basic result that domain pretraining pays. LegalBERT and CaseLaw-BERT showed consistent gains over general-domain BERT across legal classification tasks, and InLegalBERT and InCaseLawBERT extended the recipe to Indian judgments, which differ substantially in drafting convention, citation format and procedural vocabulary. These models remain the strongest task-specific specialists on several classification benchmarks and are used here both as node feature extractors and as a comparison point.

The decoder generation followed the same recipe at larger scale. SaulLM-7B is the clearest reference: a Mistral-based model given continued pretraining on roughly 30 billion tokens of legal text followed by legal instruction tuning, later scaled to 54B and 141B variants. Parallel efforts include Lawyer LLaMA, LawGPT, ChatLaw, DISC-LawLLM and InternLM-Law in the Chinese setting. LegalBench supplied the evaluation infrastructure the field lacked, with 162 tasks contributed by legal professionals and organised around a typology of legal reasoning rather than around convenient data availability.

The common property of all of this work is that adaptation happens entirely through the text channel. Structure enters only in whatever degraded form survives tokenisation.

4.2 Parameter-efficient fine-tuning

LoRA reparameterises the weight update as a product of low-rank factors, freezing the base weights and training only the factors. QLoRA combines this with 4-bit NormalFloat quantisation, double quantisation and paged optimisers, which brings 7B and 8B fine-tuning within reach of a single workstation GPU. DoRA decomposes the update into magnitude and direction and closes part of the remaining gap to full fine-tuning. These methods are not a contribution of this work; they are the enabling constraint that makes the experiment feasible at academic scale, and they define the budget that every baseline is held to so that comparisons are fair.

4.3 Graph representations of legal corpora

Legal citation networks have been studied quantitatively since long before deep learning, initially through network-analytic measures of precedent authority and influence. The relevant modern line uses these graphs as learned signal. LeSICiN is the closest direct precedent: it identifies applicable statutes for a case by combining a text encoder with a heterogeneous citation network over facts, sections and acts, and demonstrates that the structural channel adds accuracy that text alone does not provide. Complementary work on rhetorical role segmentation of judgments is what makes paragraph-level nodes semantically meaningful rather than arbitrary chunks, since a node labelled as ratio or as argument carries different weight in downstream reasoning. Argument mining over legal text supplies support and attack edges within a single document.

The limitation across this body of work is uniform. The structural signal is consumed by encoder-only classifiers, so it can support labelling but not generation, explanation or grounded summarisation.

4.4 Coupling graphs to large language models

Four families have emerged in the general graph-plus-language-model literature. The table below summarises them and states the position this project takes.

Family	Representative work	Mechanism	Assessment for this project
Linearisation	NLGraph, GraphQA prompting studies	Serialise the graph as edge lists or natural language and prompt directly	Cheap and simple, but token cost grows with graph size and accuracy degrades sharply as graphs grow. Retained as baseline B3

Family	Representative work	Mechanism	Assessment for this project
Soft graph tokens	GraphToken, GraphGPT, LLaGA, GraphPrompter	Encode with a GNN, project into the input embedding space as continuous tokens	Adopted. Fixed token cost regardless of subgraph size, and the encoder is trained through the projection
Adapter and cross-attention fusion	GraphAdapter and related designs	Inject graph state at intermediate transformer layers	Finer control but more invasive modification of the base model. Held as a secondary variant
Graph retrieval augmentation	GraphRAG, G-Retriever, HippoRAG, Think-on-Graph	Retrieve a subgraph or community summary at inference, LLM usually frozen	The PCST subgraph selection from G-Retriever is adopted for Stage 2. GraphRAG-style retrieval is retained as baseline B5

4.5 Gap analysis

Stated plainly. Legal NLP has the graphs but uses them with encoder-only models. The graph-plus-LLM literature has the fusion techniques but evaluates them on generic benchmarks such as WebQSP, ExplaGraphs and OGB citation networks. The intersection is thin.

Legal citation graphs also carry properties that generic knowledge graphs do not. There is a strict authority hierarchy, so an edge from a lower court to a superior court carries binding force while the reverse does not. Provisions have temporal validity windows, so the correct version of a provision depends on the date of the facts. Relations such as overruling and distinguishing are asymmetric and semantically loaded in a way that a generic cites edge is not. No existing graph-plus-LLM benchmark exercises any of these. Whether graph fusion helps under these conditions, and whether it specifically reduces citation hallucination, is an open question rather than a settled result.

5. Datasets and corpus plan

5.1 Task and benchmark corpora

Dataset	Jurisdiction	Scale	Task	Role in this work
LexGLUE	US, EU, ECtHR	7 tasks	Classification and multiple choice	Primary aggregate benchmark
LegalBench	Mostly US	162 tasks	Legal reasoning, mixed formats	Zero-shot and few-shot reasoning evaluation
CaseHOLD	US	53k instances	Holding identification, 5-way	Precedent reasoning; native citation edges
ECtHR Task A and B	Europe	about 11k cases	Multi-label article violation	Judgment prediction with article-level labels
SCOTUS	US	about 7.8k cases	Issue-area classification	Long-document classification
EUR-Lex and MultiEURLEX	EU	65k docs, 23 languages	Extreme multi-label over EuroVoc	Multilingual and hierarchical label structure
LEDGAR	US	80k provisions	Contract provision classification	Clause-type taxonomy
UNFAIR-ToS	EU	9.4k sentences	Unfair clause detection	Fine-grained clause-level task
CUAD	US	510 contracts, 13k spans	Clause extraction, 41 categories	Span extraction under expert annotation
ContractNLI	US	607 NDAs	Document-level NLI with evidence	Entailment plus evidence identification
MAUD	US	152 merger agreements	Expert multiple choice	Very long documents, hard reasoning
BillSum	US	22.2k bills	Summarisation	Grounded summarisation with statutory references
ILDC	India	about 35k SC cases	Judgment prediction and explanation	Indian target jurisdiction, explanation subset
IL-TUR	India	8 tasks	Multi-task Indian legal benchmark	Cross-jurisdiction transfer evaluation
AILA (FIRE)	India	Precedent and statute pairs	Retrieval	Statute and precedent retrieval
HLDC	India, Hindi	about 900k documents	Bail prediction	Low-resource and multilingual stress test
COLIEE	Japan and international	Statute and case tracks	Entailment and retrieval	Established competition benchmark

5.2 Pretraining and graph-source corpora

Source	Scale	Contribution to the graph
Pile of Law	about 256 GB across 35 sources	Domain-adaptive continued pretraining, optional stage
Caselaw Access Project	about 6.7M US decisions	Dense citation graph; primary source of cites and is-cited-by edges

Source	Scale	Contribution to the graph
US Code and CFR	Full statutory text	Containment hierarchy, definition edges, amendment history
EUR-Lex with CELEX identifiers	Full EU acquis	Machine-readable cross-references and EuroVoc concept links
Indian statutes: IPC, BNS, CrPC, BNSS, Evidence Act	Full text	Provision nodes for the Indian target jurisdiction
Indian Kanoon and Supreme Court judgments	Crawled, licence checked	Indian citation edges to pair with ILDC and IL-TUR

5.3 Graph construction methodology

Citations are extracted with a hybrid of jurisdiction-specific regular expressions, which handle standard reporter and provision formats reliably, and a fine-tuned sequence labeller for the irregular remainder, which is where most of the difficulty sits. Extracted strings are normalised and resolved to canonical node identifiers. Unresolved references are logged rather than silently dropped, because the rate of unresolved references is itself a quality signal and hiding it would make the graph look better than it is.

Paragraph nodes are produced by rhetorical role segmentation, so that each node carries a functional label such as facts, ruling by lower court, argument, statute, precedent analysis or ratio. Node features are initialised with LegalBERT or InLegalBERT embeddings of the corresponding text, which means the graph encoder begins from semantically meaningful features rather than random ones and needs fewer layers to become useful.

A held-out sample of at least 500 manually verified references is used to report extraction precision and recall per jurisdiction and per reference type. Every downstream claim depends on graph quality, so it is measured rather than assumed, and the measurement is reported whether or not it is flattering.

5.4 Splits, contamination and licensing

- Splits follow the official benchmark splits where they exist. Where a corpus is used both for graph construction and for evaluation, graph edges originating from test documents are removed before training so that the answer is not reachable through the graph.
- Contamination checks are run against the evaluation splits, since several of these benchmarks predate the base models and may appear in web-scale pretraining. Results on affected splits are reported with the caveat attached rather than quietly included.
- Only openly licensed sources are used. Provenance is documented per corpus, and anything with unclear terms is excluded rather than used on the assumption that research use is safe.

6. System architecture

The system has five stages. Stage 1 is offline and runs once per corpus. Stages 2 through 4 run per example during training and per query at inference. Stage 5 defines the training signal and the evaluation surface. Figure 1 gives the consolidated single-view overview, and Figure 2 gives the detailed data path.

6.1 Notation

Symbol	Meaning
$G = (V, E, R)$	Heterogeneous legal knowledge graph with node set V , edge set E and relation set R
G_q	Query-conditioned subgraph retrieved for input q , with at most N nodes
$H \in \mathbb{R}^{(N \times d_g)}$	Node embedding matrix produced by the graph encoder
$Z \in \mathbb{R}^{(K \times d_{\text{model}})}$	Projected graph soft tokens, K of them, in the model embedding space
$X \in \mathbb{R}^{(T \times d_{\text{model}})}$	Text token embeddings for instruction and document, length T
d_g, d_{model}	Graph encoder dimension and base model hidden dimension
a, b	Loss coefficients on the alignment and link prediction terms

6.2 Stage 1: graph construction, offline

Corpora are parsed into a heterogeneous graph with typed nodes and typed, directed relations. Node types are case document, paragraph or rhetorical unit, statutory provision, contract clause, legal concept, court, party entity and date anchor. Relation types are listed in Appendix A. Typing is what allows relation-specific message passing at Stage 3, and the ablation that collapses all relations to a single untyped edge is what tests whether the typing carries real signal or merely adds parameters.

6.3 Stage 2: query-conditioned subgraph retrieval

Passing the full graph to a GNN at every step is neither necessary nor tractable. Seed nodes are identified by dense similarity between the query embedding and node features, expanded by k -hop traversal with k typically 2, and pruned to a fixed budget of N nodes using a Prize-Collecting Steiner Tree formulation. Each node carries a prize reflecting query relevance and each edge carries a cost penalising size, so the solution is a connected subgraph that is relevant and bounded rather than a disconnected bag of high-scoring nodes.

$$G_q = \operatorname{argmax}_{\text{over subgraphs } S} \left[\sum \text{of prize}(v) \text{ for } v \text{ in } S \right] \text{ minus } \left[\sum \text{of cost}(e) \text{ for } e \text{ in } S \right]$$

N is treated as a tunable hyperparameter over the range 20 to 200. Connectivity matters here because a disconnected selection loses exactly the multi-hop paths that make the graph worth consulting in the first place.

6.4 Stage 3: heterogeneous graph encoding

The retrieved subgraph is encoded with a relation-aware GNN. The primary configuration is R-GCN, where messages are aggregated with relation-specific weight matrices:

$$h_v^{(l+1)} = \sigma \left(W_0^{(l)} h_v^{(l)} + \sum_{r \in R} \sum_{u \in N_r(v)} \left(\frac{1}{c_{vr}} \right) W_r^{(l)} h_u^{(l)} \right)$$

The alternative configuration is a Heterogeneous Graph Transformer, which replaces the fixed relation weights with type-aware attention and tends to handle heavily skewed relation frequencies better, which legal citation graphs certainly have. Depth is held at two or three layers. Deeper stacks are avoided because oversmoothing sets in quickly on citation graphs with high average degree, and because the useful signal is mostly within two

hops of the seed.

Node embeddings are pooled into K structural vectors, with K small, typically between 1 and 10. Pooling options compared are mean pooling, attention pooling conditioned on the query, and taking the K highest-prize node embeddings directly.

6.5 Stage 4: projection, injection and adaptation

A two-layer MLP with GELU activation projects the pooled vectors from d_g into d_{model} , producing K graph soft tokens that occupy the same space as ordinary token embeddings. The input sequence becomes the concatenation of the graph tokens, the instruction tokens and the document tokens.

$$\text{input} = [Z_1 \dots Z_K] \parallel [\text{instruction tokens}] \parallel [\text{document tokens}]$$

The base model is an open-source LLM in the 7B to 8B range, loaded in 4-bit and frozen. LoRA adapters are attached to the attention projections and the MLP projections with rank between 16 and 64. The trainable set is the LoRA adapters, the projector and the GNN encoder, together staying under roughly 2 percent of total parameters. Every baseline is held to the same budget, which is what makes the comparison informative rather than a demonstration that more parameters help.

6.6 Stage 5: training objective

$$L = L_{\text{task}} + a \cdot L_{\text{align}} + b \cdot L_{\text{link}}$$

Term	Definition	Purpose
L_{task}	Cross-entropy for classification and retrieval heads, next-token loss for generation	The task signal itself
L_{align}	InfoNCE contrastive loss pulling each graph token towards the text spans of the nodes it encodes	Prevents the projector from mapping graph state into a region of embedding space the model cannot interpret
L_{link}	Auxiliary citation link prediction applied to the graph encoder	Keeps the encoder structurally faithful instead of collapsing into whatever the task loss finds convenient

Coefficients a and b are tuned on a validation split, and both are subject to ablation, since an auxiliary loss that does not change results should be removed rather than defended.

6.7 Inference path

At inference the graph is fixed and cached. The query is embedded, seed nodes are retrieved by approximate nearest neighbour search over precomputed node embeddings, the subgraph is assembled and pruned, encoded, projected and prepended, and generation proceeds normally. Generated citations are checked against the graph, which is what makes the citation validity metric automatic rather than manual. Repeated queries reuse cached subgraphs.

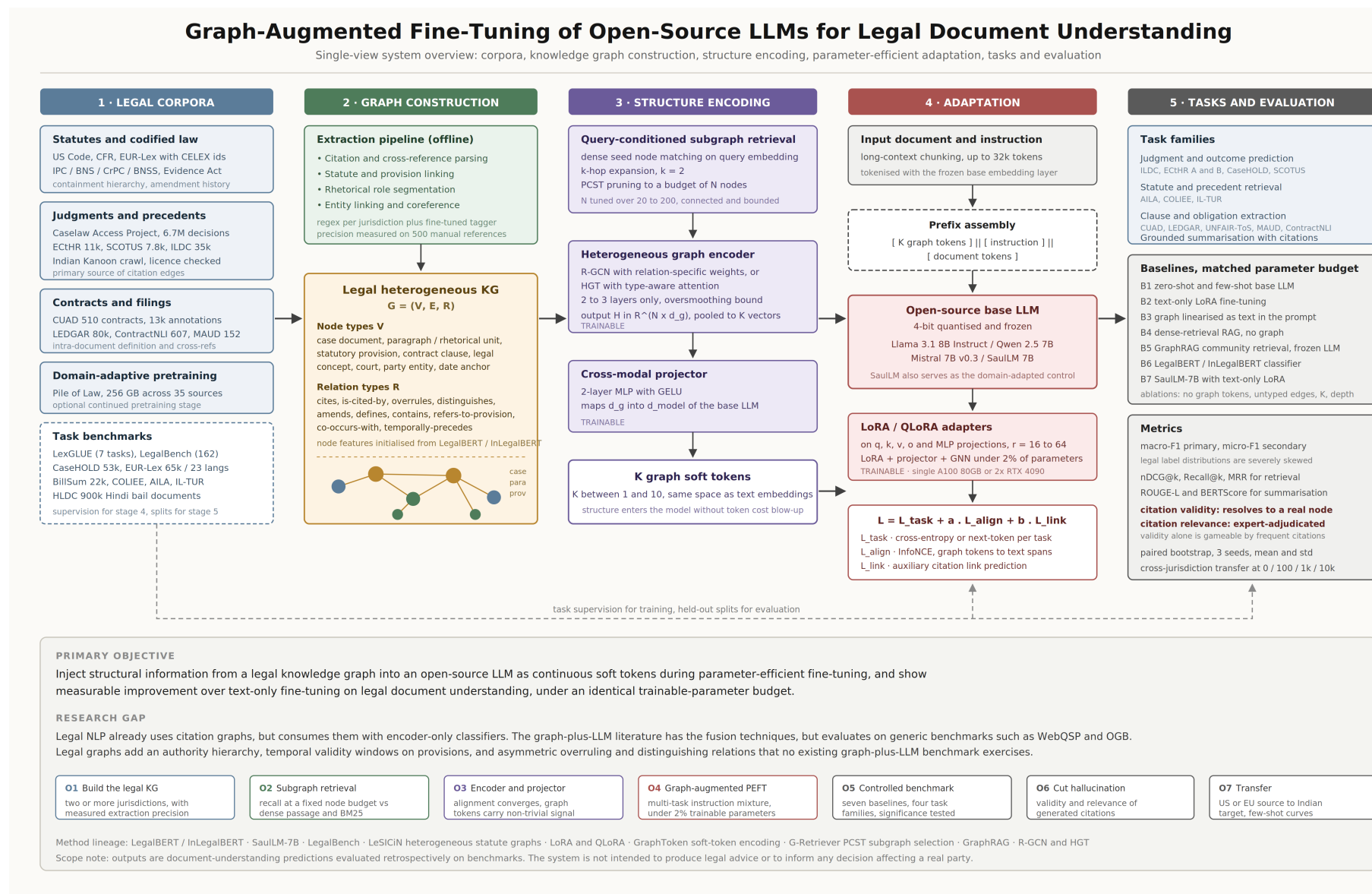


Figure 1. Consolidated single-view overview. Five stages run left to right, from corpora through graph construction, structure encoding and adaptation to tasks and evaluation. The bottom strip carries the primary objective, the research gap and the seven specific objectives, each colour-matched to the stage it belongs to. The dashed channel shows benchmark corpora supplying both training supervision and held-out evaluation splits.

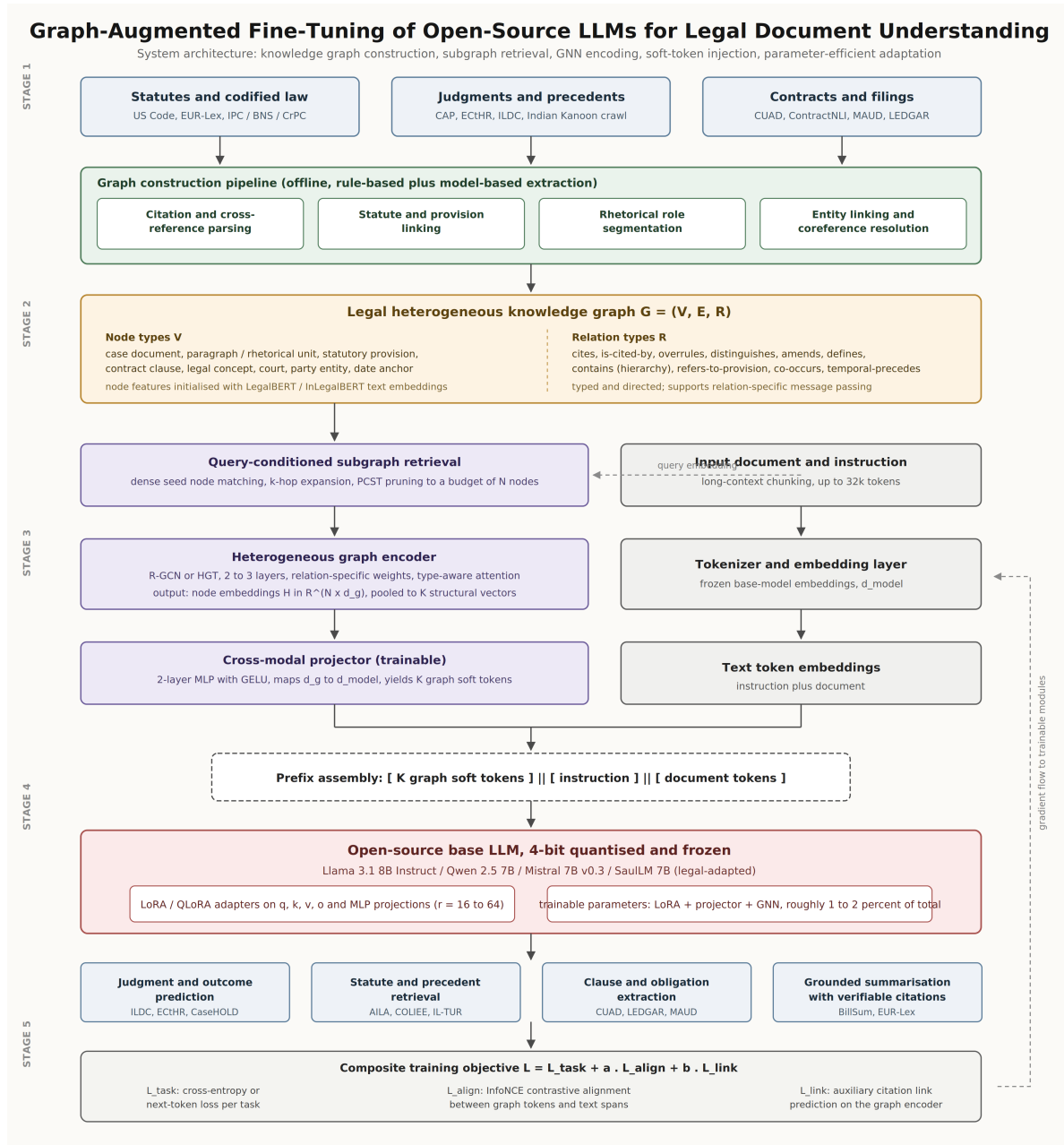


Figure 2. Detailed architecture and data path. The left branch carries structure from the knowledge graph through retrieval, encoding and projection into K soft tokens. The right branch carries the document through the frozen embedding layer. The two merge at prefix assembly and enter the quantised base model, which is adapted only through LoRA. Trainable modules are the GNN encoder, the projector and the adapters.

7. Implementation plan

7.1 Stack

Layer	Choice	Note
Base models	Llama 3.1 8B Instruct, Qwen 2.5 7B, Mistral 7B v0.3, SaulLM-7B	SaulLM doubles as the domain-adapted control for baseline B7
Training	PyTorch, Hugging Face Transformers, PEFT, bitsandbytes, TRL	4-bit NF4 quantisation, paged optimisers, gradient checkpointing
Graph	PyTorch Geometric or DGL	Heterogeneous message passing with relation-specific weights
Retrieval	FAISS with precomputed node embeddings, PCST solver	Approximate nearest neighbour for seed selection
Extraction	spaCy plus jurisdiction-specific regex, fine-tuned token classifier	Hybrid rule and model extraction with logged failures
Evaluation	vLLM for inference, custom harness over LexGLUE, LegalBench, IL-TUR	Batched generation at fixed decoding settings
Tracking	Weights and Biases or MLflow, seeded runs	Three seeds per configuration, all runs logged

7.2 Hardware envelope

One A100 80GB is sufficient for 7B to 8B training at sequence lengths up to 32k with gradient checkpointing and 4-bit quantisation. Two RTX 4090 cards with 24GB each are a workable fallback at reduced sequence length. Graph construction is CPU-bound and embarrassingly parallel, so it runs separately and is cached. The dominant cost is not the fine-tuning but the full ablation and baseline grid, which is why the grid is specified in advance in Section 8 rather than expanded opportunistically.

7.3 Default hyperparameters

Parameter	Default	Sweep range
LoRA rank r	32	16, 32, 64
LoRA alpha	64	$2r$ held fixed
LoRA dropout	0.05	0.0 to 0.1
Target modules	q, k, v, o, gate, up, down	attention only as an ablation
Learning rate	$2e-4$ for adapters, $1e-4$ for GNN and projector	$1e-4$ to $3e-4$
Batch size	4 with gradient accumulation to effective 32	hardware bound
Subgraph budget N	80	20, 40, 80, 200
Graph tokens K	5	1, 5, 10, 20
GNN depth	2	1, 2, 3
Alignment weight a	0.1	0, 0.05, 0.1, 0.5
Link weight b	0.05	0, 0.05, 0.2
Sequence length	8192 default, 32768 for long-document tasks	task dependent

8. Experimental design and evaluation

8.1 Baselines

All baselines are trained on the same data, under the same trainable-parameter budget, with the same compute ceiling and the same decoding settings. Anything else would make a favourable result uninterpretable.

ID	Baseline	What it isolates
B1	Zero-shot and few-shot base LLM, no adaptation	The floor
B2	Text-only LoRA fine-tuning	The comparison that matters most: does structure add anything at matched budget
B3	Same subgraph linearised as text in the prompt	The value of learned encoding over serialisation
B4	Dense-retrieval RAG over document chunks, no graph	The value of structure over relevance alone
B5	GraphRAG-style community retrieval, frozen LLM	Inference-time graph use without fine-tuning
B6	LegalBERT or InLegalBERT encoder classifier	The task-specific specialist ceiling on classification
B7	SaulLM-7B with text-only LoRA	Separates the graph contribution from domain adaptation

8.2 Ablations

Ablation	Question it answers
Graph tokens removed at inference only	Does the model actually use the tokens, or has it learned to ignore them
GNN replaced by mean-pooled bag of node features	Is message passing needed, or is node content sufficient
Retrieval replaced by random node sampling	Is the retrieval mechanism doing work, or is any graph context enough
Relation types collapsed to a single untyped edge	Does relation typing carry signal or only parameters
K varied over 1, 5, 10, 20	How much structural capacity is actually needed
GNN depth varied over 1, 2, 3	Where oversmoothing begins on legal citation graphs
Alignment loss removed	Is explicit alignment necessary or does the task loss suffice
Link prediction loss removed	Does structural faithfulness regularisation help
Node features randomised instead of LegalBERT initialised	How much comes from text features versus topology

8.3 Metrics

Task family	Primary metric	Secondary
Classification and judgment prediction	Macro-F1	Micro-F1, per-class F1 on rare labels
Retrieval	nDCG@k	Recall@k, MRR
Span and clause extraction	Exact match and token-level F1	AUPR at the span level
Summarisation	ROUGE-L	BERTScore, human adequacy judgment

Task family	Primary metric	Secondary
Citation faithfulness	Citation validity and citation relevance	Rate of unresolvable references per 1000 tokens

Macro-F1 is primary for classification because legal label distributions are severely skewed and micro-F1 rewards predicting the majority label. Citation validity is the fraction of generated citations that resolve to a real node in the corpus, checked automatically against the graph. Citation relevance is the fraction of valid citations that are actually pertinent to the proposition they support, adjudicated by annotators with legal training on a stratified sample. Reporting validity alone would be easy and misleading, because a model can raise it by emitting frequently occurring real citations indiscriminately.

8.4 Statistical protocol

- Every configuration is run with at least three seeds and reported as mean and standard deviation. Parameter-efficient fine-tuning at this scale is seed-sensitive enough that single run numbers are not trustworthy.
- Significance is assessed with paired bootstrap resampling over test instances, comparing against baseline B2 specifically rather than against the weakest available baseline.
- Negative results are reported. If graph augmentation fails to help on a task family, that is a finding about where structure matters and it belongs in the paper.

9. Risks and mitigations

Risk	Why it matters	Mitigation
Citation extraction errors propagate into a noisy graph	Every downstream claim rests on graph quality	Manual validation set of 500 or more references, precision and recall reported per jurisdiction, unresolved references logged rather than dropped
Graph tokens are ignored and reported gains are illusory	A model can route around an unhelpful prefix	Attention attribution over the graph token positions, plus the inference-time removal ablation
Gains come from domain adaptation rather than structure	SauLLM-style pretraining alone could explain the margin	Baseline B7 uses a legal-adapted base model, holding domain knowledge roughly constant
Subgraph retrieval becomes the inference bottleneck	A method too slow to run is not a usable method	Precomputed node embeddings, approximate nearest neighbour search, cached subgraphs, retrieval latency reported alongside accuracy
Benchmark contamination from web-scale pretraining	Several benchmarks predate the base models	Contamination checks on evaluation splits, caveats attached to affected results, preference for recent or non-public splits
Licensing restrictions on judgment corpora	Redistribution of the graph could be blocked	Openly licensed sources only, provenance documented per corpus, unclear terms excluded
Class imbalance in legal labels distorts headline numbers	Rare articles and rare clause types are the interesting cases	Macro-F1 as primary, per-class reporting on rare labels, no metric selection after seeing results
Scope creep across three jurisdictions	Breadth can dilute all three results	One jurisdiction as primary for the main table, the others confined to the transfer experiment

10. Ethics, licensing and responsible use

Legal AI carries an unusual asymmetry: a fluent wrong answer is more dangerous than an obviously incomplete one, because it is more likely to be relied upon. The measures below are part of the design rather than an appendix to it.

- **No advice framing.** All outputs are framed as document-understanding predictions on benchmark data. No deployment pathway is proposed and no claim is made about calibration sufficient for practice.
- **Citation verifiability as a first-class metric.** The system is designed so that every generated citation can be checked against the graph automatically, which is a property that text-only systems do not have.
- **Jurisdictional and demographic bias.** Case law encodes historical decision patterns. Reported performance is broken down by court and by outcome class so that a model that performs well by learning base rates is visible as such.
- **Privacy in judgment text.** Party names and personal details appear throughout published judgments. Entity nodes are retained only where they are necessary to the graph structure, and released artifacts follow the redaction conventions of the source corpus.
- **Licensing.** Only openly licensed corpora are used. The released graph carries per-source provenance so that downstream users can determine what they may redistribute.

11. Timeline and milestones

Phase	Weeks	Work	Deliverable
1	1 to 3	Corpus acquisition, licensing review, preprocessing, deduplication	Cleaned corpus with provenance manifest
2	4 to 8	Citation extraction, entity resolution, rhetorical role segmentation, graph assembly and validation	Legal knowledge graph with measured extraction quality
3	9 to 11	Subgraph retrieval implementation, PCST solver integration, recall benchmarking	Retrieval module with recall at fixed budget reported
4	12 to 15	Graph encoder and projector, alignment pretraining, sanity ablations	Aligned graph encoder checkpoint
5	16 to 19	Joint parameter-efficient training across the task mixture, hyperparameter sweep	Trained adapters for all four task families
6	20 to 23	Baselines, ablation grid, evaluation, statistical testing, error analysis	Complete result tables with significance tests
7	24 to 26	Writing, artifact preparation, documentation, release	Paper draft, code, adapter weights, graph release

11.1 Decision points

- **End of week 8.** If extraction precision on the validation sample is below an acceptable threshold for a jurisdiction, that jurisdiction is dropped from the main experiment and kept only for transfer evaluation. A weak graph is worse than no graph because it produces confident wrong structure.
- **End of week 15.** If the alignment objective does not converge or the removal ablation shows graph tokens carry no signal, the design pivots to intermediate-layer cross-attention fusion before committing the full training budget.
- **End of week 19.** The baseline and ablation grid is frozen at this point. Nothing is added after results are seen.

12. Expected contributions and deliverables

#	Contribution	Form of release
C1	A heterogeneous legal knowledge graph spanning at least two jurisdictions, with measured extraction quality and per-source provenance	Public dataset release with schema documentation
C2	A graph-augmented parameter-efficient fine-tuning architecture injecting legal structure as soft tokens	Open-source code and trained adapter weights
C3	A controlled empirical comparison against text-only PEFT, linearisation and retrieval baselines at matched budget, which the legal literature currently lacks	Full result tables with seeds, variances and significance tests
C4	A citation faithfulness protocol separating validity from relevance	Evaluation harness and annotation guidelines
C5	Cross-jurisdiction transfer results quantifying how much learned structure survives a change of legal system	Transfer curves and analysis

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Appendix A. Relation schema

Relation	Domain to range	Direction	Semantics
cites	case to case	directed	The source decision refers to the target as authority
is_cited_by	case to case	inverse	Materialised inverse, used for incoming-authority message passing
overrules	case to case	directed, asymmetric	The source displaces the target as good law
distinguishes	case to case	directed, asymmetric	The source declines to apply the target on the facts
refers_to_provision	case or clause to provision	directed	The source applies or interprets the target provision
contains	act to provision, document to paragraph	directed, hierarchical	Structural containment within a document or statute
defines	provision or clause to concept	directed	The source supplies the operative definition of the target
amends	provision to provision	directed, temporal	Carries a validity window, so the correct version depends on the date
decided_by	case to court	directed	Supports the authority hierarchy
co_occurs_with	provision to provision	undirected	Empirical co-citation, weighted by frequency
temporally_precedes	case to case	directed	Ordering used to prevent leakage of later authority

Appendix B. Node schema

Node type	Feature initialisation	Typical count order
Case document	LegalBERT or InLegalBERT embedding of headnote and ratio	10^5 to 10^6
Paragraph or rhetorical unit	Sentence embedding plus one-hot rhetorical role	10^7
Statutory provision	Embedding of provision text plus act identifier	10^4 to 10^5
Contract clause	Embedding of clause text plus LEDGAR type	10^5
Legal concept	EuroVoc or derived concept embedding	10^4
Court	Learned embedding, small vocabulary	10^2 to 10^3
Party entity	Learned embedding, retained only where structurally necessary	10^5
Date anchor	Sinusoidal or learned temporal encoding	10^4

Appendix C. Figure and artifact index

Artifact	Format	Description
Figure 1, consolidated overview	SVG and PNG	Single-view diagram of all five stages with objectives and gap statement
Figure 2, detailed architecture	SVG and PNG	Data path from corpora through graph construction to outputs and losses
Project document	PDF and Markdown	This document

End of document. Version 1.0. Bibliographic details should be verified against the publisher of record before external circulation, and all quantitative targets in Sections 8 and 11 are planning figures rather than results.